

Ye-Jun Jun

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Academic Positions

- Postdoctoral Researcher, Research Institute of Basic Sciences, Seoul National University, March 2026 to present. Weather and Climate Dynamics Laboratory. PI: Prof. Seok-Woo Son
- Visiting Researcher, Department of Geography, University of California, Santa Barbara, December 2023 to February 2024. Host: Prof. Qinghua Ding
- Research Assistant, Research Institute of Basic Sciences, Seoul National University, September 2019 to February 2020. Weather and Climate Dynamics Laboratory. PI: Prof. Seok-Woo Son

Education

- Ph.D., Atmospheric Sciences, Seoul National University, “Eurasian winter temperature trend and its decadal variability”, February 2026. Advisor: Prof. Seok-Woo Son
- M.Sc., Atmospheric Sciences, Seoul National University, “A potential driver of Eurasian winter cooling in the CESM large ensemble”, February 2022. Advisor: Prof. Seok-Woo Son
- B.Sc., Earth Science Education (major) and Atmospheric Sciences (minor), Seoul National University, August 2019

Research Interests

- Ocean–atmosphere interaction
- Tropical–extratropical teleconnections
- Atmospheric wave dynamics and wave–mean-flow interaction
- Internal variability and forced response in a changing climate
- Polar climate and Arctic–mid-latitude linkages

Modeling & Technical Skills

- Coupled climate modeling: CESM2 fully coupled simulations; implementation of SST restoring in the POP2 ocean component for regional pacemaker experiments (50-member ensembles)
- Atmospheric modeling: GFDL FMS dry dynamical core, linear baroclinic model, nonlinear stationary wave model, WRF
- Model output: single-model initial-condition large ensembles (CESM1, MPI-GE, CanESM2, GFDL SPEAR), CMIP6 (including CDRMIP)
- Observations & Reanalyses: ERA5, JRA-55, JRA-3Q, HadISST, OISST, NSIDC sea ice concentration
- Programming & Computing: Fortran, Python, HPC (Korea Meteorological Administration supercomputer)

Publications

My name in bold. * denotes corresponding author.

Peer-Reviewed Journal Articles

- 5 **Jun, Y.-J.**, S.-W. Son*, K.-H. Seo, S.-Y. Jun, D. Kim, and J.-Y. Kim (2026). Decadal reversal of Eurasian winter temperature trend linked to North Atlantic climate variability, *Geophysical Research Letters*, 53, e2026GL123035.
- 4 Cajada, M. I., S.-W. Son*, **Y.-J. Jun**, J.-H. Yoon, S.-K. Min, and S.-Y. S. Wang (2026). Re-emergence of the warm Arctic–cold Eurasia relationship by CO₂ removal, *Environmental Research Letters*, 21, 114005.
- 3 Shim, K., S.-W. Kim*, **Y.-J. Jun**, C. Park, S.-Y. Kim, S.-W. Son, D. Kim, and J.-H. Jeong (2025). East Asian dust source region restructuring linked to recent extreme drying, *Science of the Total Environment*, 1006, 180889.
- 2 **Jun, Y.-J.** and S.-W. Son* (2025). A zonal asymmetry in boreal winter surface temperature trend and its recent reversal over the Northern Hemisphere continents, *npj Climate and Atmospheric Science*, 8, 65.
- 1 **Jun, Y.-J.**, S.-W. Son*, H. Kim, H.-S. Park, and J.-H. Yoon (2024). Warm Arctic–Cold Eurasia (WACE) trend: inference from large ensemble model simulations, *Journal of Climate*, 37, 3205–3217.

Publications in the queue

- **Jun, Y.-J.**, H.-S. Park, H.-I. Jeong, C. Yoo, and S.-W. Son*. Subpolar North Atlantic warming contributes to the Eurasian winter warming hiatus, *npj Climate and Atmospheric Science*, to be submitted.
- Shim, K., **Y.-J. Jun**, H. Kim, G. Park, S.-W. Son, D. Kim, and S.-W. Kim*. Weakened relationship between springtime Mongolian cyclone activity and downward Asian dust since 2010, *Environmental Research Letters*, to be submitted.
- **Jun, Y.-J.**, S.-W. Son*, and Q. Ding. The strengthened ENSO–Arctic connection during the shoulder seasons, in preparation.
- **Jun, Y.-J.** and S.-W. Son*. Interannual variability of early winter sea ice freezing over the Barents–Kara Seas is influenced by the Quasi-Biennial Oscillation, in preparation.
- **Jun, Y.-J.**, Y. Joh*, T. L. Delworth, S.-W. Son, and co-authors. Seasonal re-coupling of Kuroshio–Oyashio Extension SST variability with the North Pacific storm track and stationary waves, in preparation.

Domestic Journal Articles (in Korean, with English abstract)

- **Jun, Y.-J.**, K. Song, and S.-W. Son* (2021). Decadal changes in the relationship between Arctic Oscillation and surface air temperature over Korea, *Atmosphere*, 31, 61–71.

Presentations

- American Meteorological Society 106th Annual Meeting, January 2026: Decadal variability of Eurasian winter temperature trend linked to North Atlantic climate variability. Oral
- American Meteorological Society 106th Annual Meeting, January 2026: A zonal asymmetry in boreal winter surface temperature trend and its recent reversal over the Northern Hemisphere continents. Poster
- Korean Meteorological Society Autumn Meeting, November 2025: Decadal change of Eurasian winter temperature trend linked to subpolar North Atlantic sea surface temperature. Oral

- Asia Oceania Geosciences Society 21st Annual Meeting, June 2024: Warm Arctic–Cold Eurasia trend in large ensemble simulations. Oral
- American Meteorological Society 104th Annual Meeting, January 2024: Interannual variability of early winter sea ice over the Barents–Kara Seas due to the Quasi-Biennial Oscillation. Poster
- Korean Meteorological Society Autumn Meeting, November 2023: Interannual variability of early winter sea ice over the Barents–Kara Seas due to the Quasi-Biennial Oscillation. Oral
- American Meteorological Society 103rd Annual Meeting, January 2023: Eurasian temperature trend in the large ensemble simulations: significant ensemble spread by tropical convection. Oral
- Korean Meteorological Society Autumn Meeting, October 2022: A potential driver of Eurasian winter cooling. Short oral
- European Geosciences Union General Assembly, May 2022: A potential driver of Eurasian winter cooling in CESM large ensemble. Short oral
- Korean Meteorological Society Autumn Meeting, October 2021: The influence of sea surface temperature over Warm Pool on the Eurasian winter temperature trend. Short oral
- Asia Oceania Geosciences Society Virtual 18th Annual Meeting, August 2021: Decadal changes in the relationship between Arctic Oscillation and surface air temperature over Korea. Poster
- Korean Meteorological Society Autumn Meeting, October 2020: Decadal changes in the relationship between Arctic Oscillation and surface air temperature over Korea. Short oral

Honors & Awards

- Merit-based Scholarship, Seoul National University, 2023
- Lecture and Research Scholarship, Seoul National University, 2022 and 2023
- BK Enhancing Diversity in Graduate Education Fellowship, Seoul National University, 2022
- Excellent Presentation Award, Korean Meteorological Society, 2021 and 2022

Teaching & Mentoring

- Research co-mentoring (with Prof. Seok-Woo Son)
 1. Jeong-Hyeon Kim, M.Sc. candidate, 2025–present, Subseasonal variability of the western North Pacific High
 2. Sung-Ah Hong, M.Sc. candidate, 2026–present, Subseasonal variability of the South Asian High
 3. Youn-Ah Kim, M.Sc., 2023–2025 (graduated February 2025). Thesis: “Compound Heatwave Events Over East Asia and Their Future Changes”
- Teaching Assistant, Large Scale Dynamics of Atmosphere (graduate course), Seoul National University, Spring 2023
- Teaching Assistant, Atmospheric Dynamics 2 (undergraduate course), Seoul National University, Fall 2022
- Teaching Assistant, Atmospheric Science Laboratory (undergraduate course), Seoul National University, Spring 2020–Spring 2022 (responsible for all lectures, lab instruction, and grading)
- Teaching Assistant, Natural Science Camp (for high school students), College of Natural Sciences, Seoul National University, 2021

Professional Activities

- Reviewer for: Nature Communications, Journal of Climate, Environmental Research Letters, Asia-Pacific Journal of Atmospheric Sciences
- Contributed translation of Chapters 4–5 for the Korean edition of C. D. Ahrens and R. Henson, *Essentials of Meteorology*, 8th ed. (Bookshill, 2021)
- Committee member (interview question development), undergraduate admissions interviews, School of Earth and Environmental Sciences, Seoul National University, 2023